

Ex.No.4 – Generate the Prompt for Different Prompt Types

Introduction

Prompt engineering is the process of designing effective instructions or questions that guide Artificial Intelligence (AI) systems to generate accurate, meaningful, and useful responses. Different types of prompts are used depending on the purpose of the task. Some prompts are designed to compare ideas, some focus on personal experiences, and others help in solving everyday problems. Understanding these prompt types improves communication between humans and AI systems.

In this experiment, three important prompt types are discussed:

1. Comparative Analysis Prompt
2. Experiential Perspective Prompt
3. Everyday Functioning Prompt

Each prompt type has a different structure and purpose. These prompts are widely used in education, business, research, customer support, and personal productivity.

1. Comparative Analysis Prompt

Definition

A Comparative Analysis Prompt is used to compare two or more items, concepts, technologies, products, or ideas. The purpose is to identify similarities, differences, advantages, disadvantages, and performance factors.

This type of prompt encourages analytical thinking and structured responses. It is commonly used in academics, product evaluation, decision-making, and research.

Features

- Compares multiple subjects
- Highlights similarities and differences
- Encourages logical reasoning
- Produces structured outputs
- Useful for decision-making

Structure of Comparative Analysis Prompt

A comparative prompt generally contains:

- Topic or subjects to compare
- Criteria for comparison
- Required format
- Expected conclusion

Example Prompt 1

“Compare the programming languages C and Python based on speed, memory usage, ease of learning, and application areas.”

Expected Response

The AI would explain:

- C is faster and memory efficient
- Python is easier to learn
- C is used in system programming

- Python is popular in AI and web development

Example Prompt 2

“Compare online learning and classroom learning in terms of flexibility, cost, interaction, and effectiveness.”

Expected Response

The AI would provide advantages and disadvantages of both learning methods and conclude which is suitable for different learners.

Applications

Comparative prompts are used in:

- Product reviews
- Educational analysis
- Technology comparison
- Market research
- Decision support systems

Advantages

- Improves analytical thinking
- Gives clear comparisons
- Helps users make decisions
- Organizes information effectively

Limitations

- Responses may become lengthy
 - Requires clear comparison criteria
 - AI may provide biased information if the prompt is unclear
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2. Experiential Perspective Prompt

Definition

An Experiential Perspective Prompt asks the AI to explain something from a personal or role-based point of view. It focuses on feelings, experiences, emotions, observations, or viewpoints.

This prompt type is useful for storytelling, empathy-based responses, creative writing, and understanding different perspectives.

Features

- Uses first-person or role-based narration
- Encourages emotional and human-like responses
- Focuses on experiences and opinions
- Useful for creative and reflective tasks

Structure of Experiential Perspective Prompt

The prompt usually contains:

- A role or perspective
- Situation or experience
- Specific details to include
- Tone or style requirement

Example Prompt 1

“Describe the experience of a student attending online classes during the pandemic.”

Expected Response

The AI may describe:

- Difficulty in concentration
- Internet connectivity issues
- Lack of social interaction
- Convenience of learning from home

Example Prompt 2

“Imagine you are a farmer facing climate change problems. Explain your daily challenges.”

Expected Response

The AI would describe:

- Unpredictable rainfall
- Crop damage
- Financial stress
- Need for modern farming techniques

Applications

Experiential prompts are used in:

- Story writing
- Education

- Psychological studies
- Training simulations
- Customer experience analysis

Advantages

- Creates engaging responses
- Encourages empathy and understanding
- Useful for creative expression
- Helps explain human experiences clearly

Limitations

- Responses may become subjective
 - AI-generated experiences may not always reflect real situations accurately
 - Requires careful wording for meaningful outputs
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3. Everyday Functioning Prompt

Definition

An Everyday Functioning Prompt helps users solve daily life problems, perform routine activities, or improve productivity. These prompts are practical and action-oriented.

They are commonly used in virtual assistants, educational systems, smart devices, and productivity applications.

Features

- Practical and task-focused
- Gives step-by-step guidance
- Easy to understand
- Supports daily activities

Structure of Everyday Functioning Prompt

These prompts generally include:

- A daily task or problem
- Required assistance
- Specific conditions or constraints
- Desired outcome

Example Prompt 1

“Create a daily study timetable for a college student preparing for exams.”

Expected Response

The AI may generate:

- Morning revision schedule
- Break timings
- Subject allocation
- Time management tips

Example Prompt 2

“Suggest healthy meal plans for a busy office worker.”

Expected Response

The AI would provide:

- Breakfast ideas
- Quick lunch options
- Healthy snacks
- Balanced dinner suggestions

Example Prompt 3

“Explain how to save electricity at home.”

Expected Response

The AI may suggest:

- Turning off unused appliances
- Using LED bulbs
- Reducing air-conditioner usage
- Energy-efficient devices

Applications

Everyday functioning prompts are used in:

- Smart assistants
- Health and fitness apps
- Educational planning
- Personal productivity tools

- Home management systems

Advantages

- Provides practical solutions
- Saves time
- Improves productivity
- Easy for users to apply

Limitations

- Responses may not suit every individual
- Requires clear instructions
- AI may miss specific personal preferences

Comparison of the Three Prompt Types

Prompt Type	Main Purpose	Style	Example Use
Comparative Analysis Prompt	Compare subjects	Analytical	Comparing C and Python
Experiential Perspective Prompt	Describe experiences	Emotional and descriptive	Student experience during online classes
Everyday Functioning Prompt	Solve daily problems	Practical and instructional	Creating study schedules

Importance of Prompt Engineering

Prompt engineering is important because AI systems depend heavily on user instructions. Well-designed prompts improve the quality, relevance, and accuracy of responses.

Benefits include:

- Better communication with AI
- Faster problem-solving
- Improved creativity
- More accurate information
- Enhanced learning experiences

Different prompt types are suitable for different purposes. Comparative prompts improve analysis, experiential prompts encourage empathy, and everyday functioning prompts support practical tasks.

Conclusion

Prompt engineering plays a major role in modern AI applications. Comparative Analysis Prompts help users evaluate and compare ideas systematically. Experiential Perspective Prompts provide human-like understanding and emotional depth. Everyday Functioning Prompts assist users in handling practical daily activities effectively.

Each prompt type serves a unique purpose and contributes to better interaction between humans and AI systems. Understanding these prompt styles helps students, researchers, and professionals create better instructions and obtain more useful AI-generated responses.

Thus, mastering different prompt types improves communication, productivity, creativity, and problem-solving abilities in the field of Artificial Intelligence.